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From Misinformation to Insight: Machine Learning Strategies for Fake News Detection

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Abstract: In the digital age, the rapid proliferation of misinformation and disinformation poses a critical challenge to societal trust and the integrity of public discourse. This study presents a comprehensive machine learning framework for fake news detection, integrating advanced natural language processing techniques and deep learning architectures. We rigorously evaluate a diverse set of detection models across multiple content types, including social media posts, news articles, and user-generated comments. Our approach systematically compares traditional machine learning classifiers (Naïve Bayes, SVMs, Random Forest) with state-of-the-art deep learning models, such as CNNs, LSTMs, and BERT, while incorporating optimized vectorization techniques, including TF-IDF, Word2Vec, and contextual embeddings. Through extensive experimentation across multiple datasets, our results demonstrate that BERT-based models consistently achieve superior performance, significantly improving detection accuracy in complex misinformation scenarios. Furthermore, we extend the evaluation beyond conventional accuracy metrics by incorporating the Matthews Correlation Coefficient (MCC) and Receiver Operating Characteristic–Area Under the Curve (ROC–AUC), ensuring a robust and interpretable assessment of model efficacy. Beyond technical advancements, we explore the ethical implications of automated misinformation detection, addressing concerns related to censorship, algorithmic bias, and the trade-off between content moderation and freedom of expression. This research not only advances the methodological landscape of fake news detection but also contributes to the broader discourse on safeguarding democratic values, media integrity, and responsible AI deployment in digital environments.

Keywords: fake news detection; machine learning; natural language processing (NLP); deep learning; text classification; computational linguistics; text mining; social media analysis; misinformation; disinformation; data analytics

Academic Editors: Jennifer D'Souza and Chengzhi Zhang

Received: 23 December 2024
Revised: 12 February 2025
Accepted: 27 February 2025
Published: 28 February 2025

Citation: Mouratidis, D.; Kanavos, A.; Kermanidis, K. From Misinformation to Insight: Machine Learning Strategies for Fake News Detection. *Information* **2025**, *16*, 189. https://doi.org/10.3390/info16030189

1. Introduction

The digital revolution has profoundly transformed the dynamics of information creation, dissemination, and consumption, accelerating the global spread of content. This transformation presents significant challenges, notably in the form of misinformation or "fake news", which includes unintentionally shared incorrect information (misinformation) and deliberately deceptive content (disinformation) [1]. Digital platforms, initially designed to enhance connectivity, have paradoxically also become battlegrounds for information integrity, impacting democratic societies significantly [2,3].

Information serves as more than a mere conduit for transmitting facts; it is a foundational pillar of social cohesion, shaping collective identities, cultural narratives, and institutional trust. In contrast, misinformation and disinformation disrupt these functions, not only by distorting truth but by undermining the structural integrity of societies. According to Cathala [4], sustained disinformation efforts do not merely manipulate perceptions—they erode the fundamental role of information in structuring human groups, potentially altering political stability and public trust. This perspective underscores that beyond individual deception, large-scale misinformation campaigns function as societal disruptions, posing threats to democratic resilience and governance structures.

Disinformation spreads six times faster than truthful information, significantly driven by emotional engagement and platform algorithms [5,6]. The decentralized nature of the internet complicates users' ability to discern relevant information, often leading to information overload and increased susceptibility to misinformation [7–10]. While misinformation typically arises from the overwhelming volume of content shared online, making it difficult for individuals to verify accuracy, disinformation is intentionally crafted to deceive and manipulate public opinion [11]. The distinction between these two concepts is critical, as misinformation often stems from a lack of verification or critical analysis, whereas disinformation exploits cognitive biases and social behaviors to maximize its reach and impact [3]. This differentiation is essential for developing appropriate countermeasures, whether through improved fact-checking mechanisms or automated detection systems. Addressing these challenges necessitates a multifaceted approach, including developing advanced artificial intelligence tools for automated detection of disinformation [12] and implementing legislative measures to protect citizens against deceptive content [13].

Beyond misinformation and disinformation, the concept of mal-information has emerged as a critical distinction in the study of digital information integrity. Unlike misinformation, which results from unintentional errors, and disinformation, which is deliberately false, mal-information refers to true information that is used with the explicit intent to harm an individual, organization, or country [11]. Examples include leaked confidential documents, selectively edited content to misrepresent context, or exposing private data to damage reputations. Mal-information often exploits factual accuracy while distorting intent, making it particularly challenging to address using automated detection systems [14]. This distinction is crucial in defining the broader landscape of harmful information practices and underscores the importance of context-aware detection models that go beyond the binary classification of truthfulness.

Historically, misinformation has been utilized for manipulation—from its role in propaganda during World War I to its strategic use in early corporate public relations [15]. The advent of the digital era, characterized by instantaneous communication and minimal regulatory oversight, has dramatically expanded the reach and impact of misinformation. Social media platforms exacerbate this issue by prioritizing user engagement over content accuracy, thus facilitating the rapid spread of potentially harmful misinformation [14]. Additionally, misinformation is amplified through information cascades, where users unknowingly share misleading content without verifying its authenticity. Studies have shown that falsehoods are more likely to be shared than factual information, as they often evoke stronger emotional reactions, leading to viral dissemination [6]. This highlights the importance of designing detection systems that not only identify falsehoods but also account for the mechanisms through which misinformation spreads.

Economic and ideological motives primarily drive the proliferation of fake news. Economically motivated fake news seeks to generate revenue through sensational content that attracts clicks, while ideologically driven misinformation aims to manipulate public opinion and deepen societal divisions [16]. The significant influence of such content was starkly observed during the 2016 US presidential elections, where misinformation campaigns notably swayed public sentiment and behavior, as seen in the "Pizzagate" incident [17]. The economic incentives behind misinformation are further exacerbated by digital advertising models, where virality is rewarded, even at the cost of factual accuracy [9]. Platforms that rely on engagement-driven revenue models contribute to the spread of misleading narratives, making it imperative to integrate detection methodologies that consider not just the textual properties of fake news but also its dissemination patterns.

Beyond political disruption, the consequences of misinformation affect economic stability and public health, with false information potentially triggering stock market fluctuations and public health crises. Addressing these multifaceted threats necessitates a holistic approach encompassing technological innovation, media literacy, and adaptable regulatory frameworks.

This research aims to develop a robust and explainable framework for fake news detection by leveraging diverse machine learning architectures and vectorization techniques. The key objectives and contributions of this study include:

- Advancing Text Representation Techniques for Fake News Detection: We assess the effectiveness of various text vectorization methods—including TF-IDF, Bag of Words (BoW), Word2Vec, and BERT embeddings—highlighting their impact on classification accuracy and computational efficiency. Our findings confirm that contextual embeddings (BERT) significantly enhance fake news detection compared to traditional feature extraction techniques.
- Comprehensive Performance Benchmarking of Machine Learning and Deep Learning Models: We systematically evaluate classical classifiers (Naïve Bayes, SVM, Random Forest) and deep learning models (CNNs, LSTMs, BERT) across multiple fake news datasets. This comparative analysis identifies the most effective architectures, demonstrating that BERT consistently outperforms alternative approaches in terms of F1 score and MCC.
- Addressing Data Imbalance and Improving Model Fairness: To mitigate biases introduced by imbalanced datasets, we apply the Synthetic Minority Over-sampling Technique (SMOTE) and class-weighted loss functions. Our experiments demonstrate that SMOTE enhances model robustness and improves classification metrics, particularly for underrepresented fake news categories.
- Explainability and Model Interpretability for Fake News Detection: We integrate Explainable AI (XAI) techniques such as SHAP and LIME to analyze model decision-making. Feature importance analysis and attention heatmaps for BERT provide transparency, ensuring interpretability and trustworthiness in automated misinformation detection, particularly in high-stakes scenarios.
- Extensive Evaluation with Multiple Performance Metrics: Beyond accuracy and F1 score, we introduce the Matthews Correlation Coefficient (MCC) and ROC–AUC to provide a more holistic evaluation of model performance. The results confirm that MCC offers a more reliable assessment in imbalanced datasets, reinforcing our methodological choices.
- Ethical Considerations and Practical Implications: We discuss the ethical challenges of automated misinformation detection, including trade-offs between censorship, user privacy, and content moderation responsibilities. Our findings emphasize the need for hybrid detection frameworks that balance technical performance with ethical constraints, ensuring responsible AI deployment.

Additionally, we examine the ethical implications and potential biases of automated detection systems, addressing the balance between censorship and freedom of expression, platform responsibilities in content moderation, and user privacy concerns. A key challenge in misinformation detection lies in distinguishing harmful disinformation from satirical or opinion-based content. Our research considers these complexities and evaluates the broader impact of detection models on digital discourse [18,19]. This research propels technological solutions forward and contributes to the discourse on preserving democratic values in a digitally interconnected era, promoting enhanced digital literacy and media integrity.

The remainder of this paper is structured as follows: Section 2 reviews the existing literature, highlighting previous efforts and methodologies in fake news detection, particularly focusing on BERT-based models and their comparative effectiveness. Section 3 details our methodological framework for detecting fake news, describing the integration of traditional machine learning models, deep learning architectures, and advanced text vectorization techniques. Section 4 presents the computational environment and datasets used, ensuring reproducibility of the results. Section 5 provides an extensive analysis of our models' performance across multiple datasets, incorporating SMOTE-based evaluation for addressing data imbalance, SHAP and LIME explainability methods for model transparency, and alternative evaluation metrics such as MCC and ROC–AUC to ensure a comprehensive assessment. Finally, Section 6 summarizes our key findings, discusses the ethical and practical implications of automated misinformation detection, and outlines future research directions aimed at enhancing detection frameworks for real-world deployment.

2. Related Work

The digital communication revolution has exponentially increased the dissemination of misinformation, compelling the development of sophisticated detection strategies. This section delineates the evolution of methodologies for detecting fake news, tracing the transition from early manual and rule-based approaches to the advanced realms of machine learning and deep learning. This progression reflects the growing complexity and subtlety of misinformation in the digital age.

Initial studies in fake news detection predominantly employed traditional machine learning techniques, emphasizing the critical role of feature selection [20–23]. Incorporating a blend of linguistic cues, network-based features, and meta-information has been shown to significantly augment the detection capabilities of these systems [24]. Analysis of traditional classifiers such as Decision Trees, Random Forests, and Support Vector Machines reveals that ensemble methods, which amalgamate insights from multiple models, are especially effective in enhancing prediction accuracy and mitigating overfitting [25,26].

With the advent of deep learning, the scope of fake news detection has expanded to include more nuanced textual and contextual analyses [27]. Cross-lingual transfer learning, for instance, has facilitated the application of models across diverse linguistic landscapes, markedly boosting performance in multilingual settings [28,29]. Moreover, the advent of sophisticated models such as BERT has proven highly effective in parsing the intricacies of context and semantics in textual content, significantly refining the accuracy of detection systems [30].

BERT-based approaches have been widely explored for misinformation detection due to their strong contextual understanding and language modeling capabilities. Studies such as [31,32] have demonstrated the effectiveness of fine-tuned BERT models in classifying fake news across different datasets. Additionally, hybrid models that combine BERT with CNNs and LSTMs have been proposed to enhance feature extraction, capturing both semantic and temporal dependencies [33,34]. These approaches highlight BERT's versatility in handling textual misinformation across domains, from social media posts to full-length news articles. However, prior studies have primarily focused on fine-tuning single BERT models without systematically integrating complementary vectorization techniques, preprocessing refinements, or ensemble learning frameworks to further improve robustness.

Despite these advancements, limitations remain in existing BERT-based solutions. Many studies rely on static datasets and fail to evaluate model generalization across multiple sources and evolving misinformation patterns [35]. Moreover, previous research has primarily benchmarked BERT against traditional ML classifiers but lacks extensive performance comparisons against other Transformer-based architectures or hybrid deep learning techniques (e.g., BERT-CNN, BERT-LSTM, knowledge graph-enhanced models). While some works have examined fine-tuning strategies for BERT, fewer have systematically evaluated the combined impact of advanced feature engineering, preprocessing optimizations, and multi-level ensemble learning for misinformation detection. Our work addresses these gaps by integrating a refined preprocessing pipeline, diverse vectorization techniques, and a comparative evaluation of BERT against traditional and hybrid deep learning models. Additionally, we conduct an extensive multi-dataset evaluation to ensure the robustness and adaptability of our framework, providing deeper insights into the scalability of BERT-based approaches in real-world misinformation detection scenarios.

Recent advancements have underscored the efficacy of hybrid models, which meld various machine learning methodologies to create robust detection frameworks. Such models, integrating textual, visual, and contextual data, embody a holistic approach to tackling the complex nature of fake news [36,37]. Integrating diverse techniques enhances adaptability and accuracy, setting new benchmarks in fake news detection strategies.

Innovations continue developing hybrid deep learning models that synergize Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM) networks. This combination leverages CNNs' proficiency in extracting spatial patterns and LSTMs' capability in sequence prediction, crafting a potent analytical framework suitable for fake news content's dynamic and complex nature [38].

Additionally, specialized models like the exBAKE, built upon the BERT architecture, have been tailored to meet the unique challenges of fake news detection. Such models achieve high accuracy levels by adeptly interpreting the context and semantics embedded within news articles, showcasing the potential of fine-tuned deep learning architectures in this field [30,39].

As fake news detection methodologies evolve, it becomes crucial to address their ethical implications. Concerns regarding algorithmic bias, privacy, and the automation of content moderation pose significant challenges that require careful consideration and ongoing research.

The convergence of traditional machine learning algorithms with cutting-edge deep learning techniques forms a promising frontier in combating misinformation. This integration not only tackles the inherent challenges posed by the dynamic nature of social media content but also keeps pace with continuous technological innovations, enhancing the scalability and robustness of detection systems.

In summary, the field of fake news detection is characterized by rapid evolution, propelled by technological advances and the increasingly sophisticated nature of misinformation. Ongoing research and development are critical in devising innovative solutions that can swiftly adapt to the changing landscape of digital information dissemination. As methodologies evolve, they lay the groundwork for future initiatives to preserve the integrity of information on digital platforms, thereby safeguarding democratic values and public discourse.

3. Methodological Framework for Fake News Detection

The rapid proliferation of digital media has escalated the spread of fake news, posing significant challenges to information credibility and public trust. To address this, our research develops a robust methodological framework to detect fake news effectively across various digital platforms. This framework integrates advanced data preprocessing, sophisticated feature extraction techniques, and a hybrid machine learning and deep learning model approach. Each component is meticulously crafted to enhance detection accuracy and adaptability, ensuring our methodology remains effective against evolving misinformation tactics.

3.1. System Overview

Our fake news detection system is a multi-component pipeline that processes data from input to output, transforming raw data into actionable insights. The system comprises the following main components:

- Data Preprocessing: The initial stage where raw data are cleaned and prepared for analysis.
- Feature Extraction: Techniques are applied to transform preprocessed data into a feature-rich format suitable for machine learning models.
- Model Training: Application of machine learning and deep learning algorithms to train models on features derived from the data.
- Model Evaluation: Techniques used to assess the accuracy and efficacy of the trained models.
- Decision Making: Integration of model outputs to make final determinations on the veracity of news items.

3.2. Data Preprocessing

Effective fake news detection begins with meticulous data preparation. The preprocessing stage is crucial for enhancing the data quality and ensuring its suitability for complex analyses. This phase involves several key processes designed to normalize and refine the text data, which include:

- Text Cleaning: We remove HTML tags, URLs, special characters, and any extraneous whitespace. This standardization simplifies the textual content, making it more amenable to analysis.
- Noise Reduction: Common stop words (e.g., "and", "the", "of") are removed, and non-informative filler words are filtered out to focus on significant text elements. This step reduces dataset noise and enhances the salience of meaningful content.
- Normalization Techniques: Text is tokenized into words or phrases, and normalization techniques such as stemming and lemmatization are applied. Stemming truncates words to their root forms, while lemmatization converts words to their dictionary form, taking into account their part-of-speech and contextual meanings. This is facilitated using tools like the NLTK Snowball Stemmer and WordNet Lemmatizer [40].

To address potential biases in our datasets, especially due to class imbalances, we employ techniques such as SMOTE for synthetic data generation and adjust classifier thresholds to balance accuracy and recall. These steps are aimed at ensuring our model's fairness and effectiveness across varied scenarios.

In addition to employing SMOTE for synthetic data generation to balance underrepresented classes, we also experiment with class-weighted loss functions in our deep learning models. This method assigns higher weights to the minority class during training, ensuring that the model does not disproportionately favor the majority class. This approach is particularly beneficial for deep learning architectures where oversampling may lead to redundant information rather than true diversity in the training data.

3.3. Feature Extraction and Representation

Following data preprocessing, feature extraction is the subsequent pivotal phase in our methodology. This critical stage transforms the normalized text into structured data formats amenable to detailed analysis by machine learning algorithms. We utilize diverse vectorization techniques adept at capturing different linguistic and semantic data characteristics, providing a robust basis for subsequent classification tasks.

- Vectorization Techniques:
  - TF-IDF (Term Frequency-Inverse Document Frequency): This technique quantifies the importance of each word within the documents relative to its frequency across the entire corpus. It is beneficial for filtering out common words that are less informative in the corpus context. The TF-IDF value increases proportionally with the number of times a word appears in the document. However, it is offset by the frequency of the word in the corpus, which helps to adjust for the fact that some words appear more frequently in general. TF-IDF is mathematically represented as:
  $$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t) \tag{1}$$
  where TF(t, d) is the term frequency, and IDF(t) is defined as:
  $$\text{IDF}(t) = \log\left(\frac{N}{n_t}\right) \tag{2}$$
  with N being the total number of documents and n_t the number of documents containing the term t. This formula helps diminish the weight of terms that occur very frequently in the document set and increase the weight of terms that occur rarely.
  - Bag of Words (BoW): Constructs a simple model where the presence of words in a document is noted. However, the order does not simplify the problem of document classification. The BoW model converts text documents into a fixed-length vector of word counts. The vector for a document d is represented as:
  $$\vec{v}(d) = (c_1, c_2, \ldots, c_M) \tag{3}$$
  where M is the total number of unique words in the corpus, and c_i is the count of word i in document d. This model is adequate for datasets with manageable vocabulary, allowing straightforward frequency-based classification of texts.
  - Doc2Vec: An extension of the Word2Vec methodology, Doc2Vec considers the semantic meaning of individual words and maintains a unique vector for each document, which captures the context of the entire text. This method is advantageous for content-rich datasets where the contextual narrative plays a crucial role in determining the authenticity of the news.
  - BERT Embeddings: Employs the Transformer architecture to generate deep contextual representations for every word in the text based on the other words in the sentence. Unlike earlier methods that generate a single word embedding for each word in the vocabulary, BERT considers the full context of a word by looking at the words that come before and after it, making it exceedingly effective for understanding the complex structures within sophisticated fake news stories.
- Additional Semantic Features:
  - Average Sentiment Polarity: This metric assesses the overall sentiment of the text, quantifying it as positive, neutral, or negative. Emotional analysis is crucial for detecting fake news, which often exploits strong emotional cues to mislead readers. The polarity score provides insights into the emotional tone of the content, which, combined with other features, helps distinguish manipulative news articles [41].

– Readability Score (Flesch–Kincaid): The readability of an article can offer subtle cues about its authenticity. Highly complex or overly simplistic texts can be red flags for misinformation. The Flesch–Kincaid readability test quantifies how easy it is to understand the text based on word length and sentence complexity. This score is particularly useful for identifying texts that may be crafted to deceive or manipulate readers with complex jargon or overly simplified language [42].
– Thematic Diversity: Analyzing the diversity of topics covered in texts helps understand the spread and focus of content, which can indicate fake news. This feature looks at how specific themes are represented across different datasets, providing a way to identify potential biases or the targeting of specific audiences with disinformation [43].

By integrating these advanced vectorization and semantic analysis techniques, our methodology is equipped to handle the multifaceted nature of fake news, capturing both overt and subtle indicators of falsehood. This detailed approach ensures that our models are not only sensitive to the explicit features of the text but also to the deeper, more nuanced aspects of language that characterize deceptive information.

3.4. Algorithmic Framework

In our approach to detecting fake news, we leverage various machine learning algorithms, each chosen for its unique strengths and suitability for different aspects of the problem. This subsection details each algorithm used, exploring its theoretical basis, application in our context, and its specific role in enhancing the detection system's performance.

- Naïve Bayes: This algorithm applies Bayes' theorem with the "naïve" assumption of conditional independence between every pair of features given the value of the class variable. Naïve Bayes is particularly effective for large datasets and can be implemented with:
$$P(c|x) = \frac{P(x|c)P(c)}{P(x)} \tag{4}$$
where $P(c|x)$ is the posterior probability of class c given predictor x, $P(c)$ is the probability of class c, and $P(x|c)$ is the likelihood which is the probability of predictor given class.
- Support Vector Machines (SVM): SVMs are powerful for classification problems and work well on both linear and non-linear data. In the context of fake news, SVMs help us effectively segregate genuine news from fake news, especially when the differentiation is not immediately apparent from the content's features. They attempt to find the hyperplane that best separates different classes by maximizing the margin between the closest vectors of the classes (support vectors), defined as:
$$f(x) = \text{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b\right) \tag{5}$$
where $K(x_i, x)$ represents a kernel function which transforms the input data into a higher-dimensional space.
- Random Forest: As an ensemble learning method, Random Forest uses multiple decision trees to improve classification accuracy. Each tree in the forest is built from a sample drawn with replacement from the training set. Averaging or 'voting' from the results of different trees reduces overfitting and is less sensitive to outliers. This

method is invaluable when dealing with fake news articles' heterogeneous and often deceptive features. The general formula for prediction in classification problems is:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_n(x)\} \tag{6}$$

where $T_i(x)$ are the individual tree predictions, and mode is the statistical mode of these predictions.

- Convolutional Neural Networks (CNNs): Originally designed for image processing, CNNs have been adapted for NLP tasks due to their ability to pick out invariant features. For text, CNNs can effectively recognize patterns within a block of text, such as phrases or sentences, that have a high predictive power for classification. This ability makes them excellent at identifying stylistic features and contextual cues in fake news content. For text, a typical convolutional operation might look like:

$$c_i = \text{ReLU}\left(b + \sum_{m=1}^{M} x_{i+m-1} w_m\right) \tag{7}$$

where c_i is the feature map output, x_{i+m-1} are the input words or characters, w_m are the weights of the filter at position m, b is a bias term, and ReLU is the activation function.

- Long Short-Term Memory Networks (LSTMs): LSTMs are particularly adept at processing data sequences, making them ideal for analyzing text flow in news articles. They can model the temporal sequence of statements to identify patterns that are indicative of fake news, such as inconsistencies or exaggerations that develop throughout the text. The mathematical backbone of LSTMs involves gates that regulate the flow of information, retaining or forgetting details through structures called cell states and hidden states:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{8}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{9}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{10}$$

$$c_t = f_t * c_{t-1} + i_t * \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{11}$$

$$h_t = o_t * \tanh(c_t) \tag{12}$$

where σ denotes the sigmoid activation function, and W and b represent the weights and biases for the forget gate (f), input gate (i), output gate (o), and cell candidate (c), respectively.

- BERT (Bidirectional Encoder Representations from Transformers): BERT's ability to analyze the context of words from both left and right in a sentence all at once allows it to capture the deeper meanings of words based on their usage across different contexts. This feature is pivotal in distinguishing between genuine news and sophisticated fake narratives that might exploit ambiguous or nuanced language to mislead readers. The model is pre-trained on a large corpus and fine-tuned for specific tasks like fake news detection using the following Transformer architecture formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{13}$$

where Q, K, and V represent the query, key, and value matrices derived from the input data, and d_k is the dimension of the keys.

Each algorithm is rigorously tested and optimized to ensure it performs at its peak within the integrated system. The choice and deployment of these algorithms are based on their proven strengths in various text analysis scenarios, providing a robust framework that enhances our ability to detect and classify fake news accurately.

3.5. System Architecture and Workflow

Figure 1 presents the comprehensive architecture of our fake news detection system, emphasizing the sequence of operations from raw data input to the final decision-making process. This visual depiction clarifies how each system stage interconnects and contributes to the ultimate goal of accurately classifying news articles.

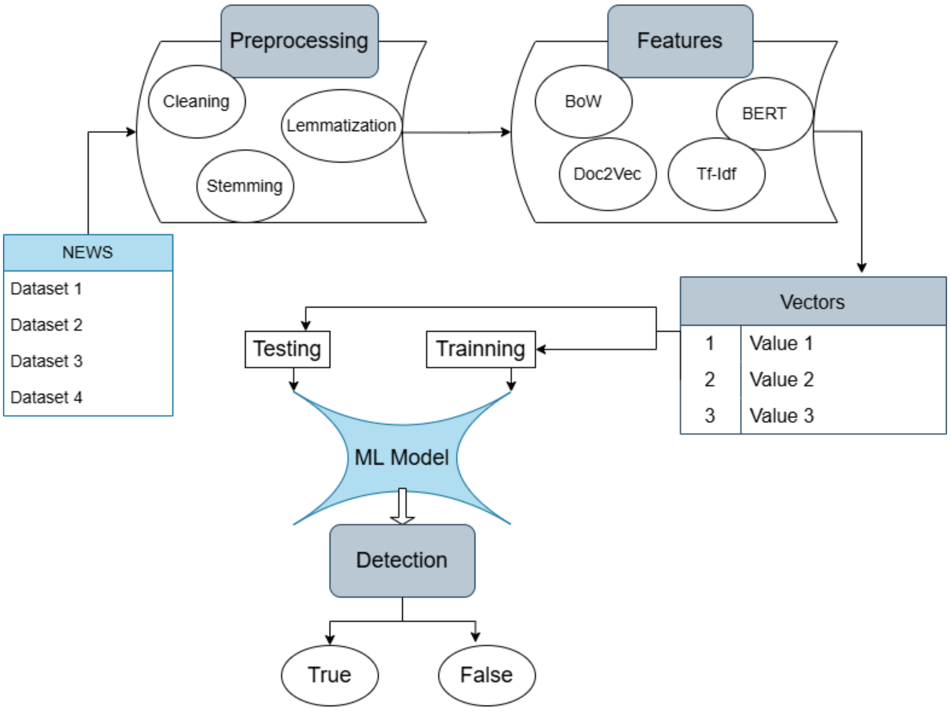

Figure 1. Comparative architecture and performance of machine learning models.

The architecture begins with the preprocessing phase, where raw news data undergo several cleaning and filtering processes to ensure the data are usable for feature extraction. These processes include removing irrelevant characters, filtering out noise, and standardizing the text data. This stage is critical as it directly impacts the quality and effectiveness of the feature extraction phase.

After preprocessing, the system extracts various features from the cleaned text. These features encompass a broad range of data points, from basic text attributes to complex linguistic and semantic markers. The extracted features are then transformed into numerical vectors using different vectorization techniques such as Bag of Words (BoW), Doc2Vec, BERT, and TF-IDF. Each vectorization method has its strengths and is chosen based on the specific requirements of the dataset and the desired accuracy.

These vectors serve as input for the machine learning models in the training phase. The dataset is divided into training and testing sets to evaluate the models' performance and to ensure they are not overfitting but are generalizing well on new, unseen data. The ML models include a mix of traditional algorithms like SVM and Random Forest and advanced deep learning models like CNNs and LSTMs, which are trained to identify patterns and anomalies indicative of fake news.

The decision-making process at the system's output stage categorizes news articles as 'True' or 'False'. This binary classification is the culmination of the entire process, where the models' predictions based on the learned patterns in the training phase are applied to real-world data to detect fake news.

3.6. Model Training

Model training is a critical phase where theoretical models are applied to practical data. This subsection describes the training process, including how the data are partitioned, the optimization of model parameters, and specific strategies employed to enhance model effectiveness:

- Data Splitting: The dataset is divided into training, validation, and test sets. Typically, the data are split into 70% training, 15% validation, and 15% test sets, ensuring that models are trained and rigorously tested on unseen data to evaluate their generalizability.
- Parameter Tuning: Using techniques such as grid search and random search, we optimize the hyperparameters of each model. This step is crucial to find the most effective settings for algorithms, such as the learning rate, number of layers, or tree depth, depending on the model in use.
- Training Techniques: Advanced methods such as batch normalization, dropout for neural networks, and ensemble techniques for decision trees are employed to prevent overfitting and ensure robust performance across diverse data scenarios.

By systematically addressing these components, our training process is designed to maximize the predictive performance of our models, ensuring they are both accurate and adaptable to new, unseen datasets.

3.7. Model Evaluation and Validation

After training, models undergo a thorough evaluation process to assess their effectiveness in detecting fake news. This subsection details the evaluation metrics, validation techniques, and overall performance assessment strategies:

- Performance Metrics: We utilize a variety of metrics to assess model performance, including accuracy, precision, recall, and the F1 score. These metrics provide a comprehensive view of model efficacy in classifying news correctly.
- Cross-Validation: To ensure the models perform well across different subsets of the dataset, k-fold cross-validation is employed. This technique enhances the reliability of performance estimates, helping to mitigate any potential overfitting.
- Validation Techniques: Models are validated using an independent test set, which was not used during the training phase. This practice confirms that the models have generalized well beyond the training data.

The meticulous evaluation and validation ensure that our models are theoretically sound and practically practical, providing robust tools for identifying fake news in real-world applications.

3.8. Experimental Setup

The experiments are designed to rigorously test each model's performance across multiple datasets. This testing not only provides insights into the models' accuracy and efficiency but also helps understand the impact of different preprocessing and feature extraction techniques on overall system performance.

By systematically addressing each component of the fake news detection pipeline, from data preparation to model evaluation, this methodology ensures a thorough and practical examination of various strategies to combat misinformation. Integrating multiple datasets and diverse algorithms offers a comprehensive perspective on the challenges and potentials of current approaches in the field.

3.9. Model Interpretability

Understanding how our models make decisions is crucial, particularly in the sensitive context of fake news detection. Interpretability ensures transparency and builds trust in automated systems, particularly in high-stakes scenarios such as political misinformation or public health-related fake news. To address this need, we employ multiple explainability techniques to provide insights into how our models arrive at their decisions.

For traditional tree-based models like Random Forest and XGBoost, we utilize feature importance analysis, which quantifies the contribution of each input feature to the model's classification decisions. This helps highlight the most influential linguistic, structural, or contextual indicators in determining the veracity of news articles.

For deep learning models, particularly BERT, we employ attention visualizations, which provide insight into which words or phrases the model focuses on when making a classification decision. By analyzing attention heatmaps, we can determine whether BERT assigns higher importance to specific misleading phrases, sentiment-heavy words, or contextual relationships indicative of misinformation.

To further enhance interpretability, we incorporate SHapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). These techniques allow us to analyze individual classification instances, providing detailed explanations for each prediction. SHAP values quantify the contribution of each feature to a given prediction, ensuring that stakeholders can understand how specific textual elements influence the final classification. Meanwhile, LIME approximates complex models by generating local interpretable surrogate models, offering transparency even for high-dimensional deep learning-based approaches.

By integrating these techniques, we ensure that our framework is not only effective in detecting misinformation but also interpretable and accountable. These explainability mechanisms facilitate model auditing, allowing researchers, policymakers, and fact-checkers to better understand how automated systems classify fake news, improving trust and adoption of AI-driven solutions in real-world applications.

4. Implementation

This section provides a detailed description of the practical implementation of the methodologies outlined in the previous sections, including specifics about the datasets used, the computational setup, and the execution of machine learning and deep learning models.

4.1. Evaluation Metrics Overview

As part of our implementation strategy, we employ rigorous evaluation metrics to assess the effectiveness of our models:

- Val F1: This metric calculates the F1 score on the validation set for each fold in our cross-validation process. It provides critical insights into the model's ability to generalize, indicating how well it performs on unseen data after training on other data segments.
- Mean F1: This metric represents the average of the Val F1 scores across all folds in the cross-validation. It is a robust and comprehensive measure of overall model performance, aggregating results to minimize the impact of any single-fold anomalies.

These metrics are pivotal in our experimental setup, allowing us to gauge both the individual and aggregate effectiveness of our models across different segments of data, thereby highlighting their predictive power and stability.

4.2. Datasets

Our study uses four distinct datasets to assess the performance of fake news detection models, each varying in size, content, and linguistic features as presented in Table 1:

- Dataset 1: Titled "George McIntyre—Fake News", contains 6355 articles (60% true, 40% false), with an average length of 500 words, slightly negative sentiment, and moderate readability (grade 8) [44]. This dataset's small size and focus on politics and social issues may limit model generalization.
- Dataset 2: From the UTK ML Club Kaggle competition, includes 26,000 articles (55% true, 45% false), averaging 600 words with neutral sentiment and easy readability (grade 6) [45]. It offers a broader range of topics, though class imbalance is a potential concern.
- Dataset 3: The ISOT Fake News dataset holds 44,900 articles (50% true, 50% false), averaging 700 words with neutral sentiment and moderate readability (grade 8), covering diverse topics like politics and technology [46]. Its large size allows robust model training but requires significant computational resources.
- Dataset 4: Combines 20,000 articles of Getting Real About Fake News dataset from Kaggle and the Signalmedia One Million News Articles dataset (65% true, 35% false) from multiple sources, with neutral sentiment, moderate readability (grade 7), and a focus on politics and international relations. However, blending sources may introduce labelling inconsistencies [47].

Table 1. Summary of datasets used for fake news detection.

Characteristic	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
Total Articles	6355	26,000	44,900	20,000
True Segments (%)	60	55	50	65
False Segments (%)	40	45	50	35
Average Article Length (words)	500	600	700	550
Average Sentiment Polarity	Slightly negative (−0.2)	Neutral (0.05)	Neutral (0.1)	Neutral (0.05)
Readability Score (Flesch–Kincaid)	Moderate (Grade 8)	Easy (Grade 6)	Moderate (Grade 8)	Moderate (Grade 7)
Thematic Diversity (Top Topics)	Politics, Social Issues	Politics, Economy, Health	Politics, Technology, Entertainment	Politics, International Relations
Key Features	Small dataset, specialized articles, limited diversity	Balanced dataset, mix of real and fake news	Large-scale dataset, well-balanced, high computational resource requirements	Diverse dataset, combination of sources, blending concerns

4.3. Computational Resources

The experiments are conducted on a high-performance computational setup, which includes:

- Hardware: NVIDIA GPUs with Tensor cores (NVIDIA, Santa Clara, CA, USA), 64 GB of RAM, essential for handling large datasets and running complex deep learning models efficiently.

- Software: The software stack is built on Python 3.13-based libraries such as scikit-learn, TensorFlow 2.16.1, and PyTorch 2.5.0, which are used for efficient model development and experimentation.

This section thoroughly details each aspect of the implementation, from dataset characteristics to the computational environment and the evaluation metrics used. This ensures that the research is both reproducible and transparent, critical for validating the study's results and facilitating future research based on these findings.

5. Experimental Results

This section presents a comprehensive analysis of the performance of various machine learning models and vectorization techniques in the context of fake news detection. We employ cross-validation techniques to ensure robust evaluation across multiple data folds, focusing on the F1 score, a critical metric for assessing model performance in imbalanced datasets. The F1 score, which is the harmonic mean of precision and recall, is particularly valuable for understanding how effectively each model balances the identification of true positives against the risk of false positives and false negatives, which is crucial in the domain of misinformation where both types of errors have significant consequences.

To ensure a detailed assessment, we extend our evaluation beyond traditional performance metrics by incorporating Matthews Correlation Coefficient (MCC) and Receiver Operating Characteristic–Area Under the Curve (ROC–AUC). These metrics offer a more nuanced understanding of model effectiveness, particularly in imbalanced settings where accuracy alone may be misleading. MCC integrates true positives, true negatives, false positives, and false negatives into a single metric, making it particularly valuable for determining model reliability. Similarly, ROC–AUC provides a broader measure of a model's discrimination power across classification thresholds, capturing its ability to separate real and fake news effectively.

In addition to quantitative performance metrics, this section explores the qualitative aspects of model behavior under different preprocessing and configuration settings, shedding light on the adaptability and efficiency of each approach across various content types and linguistic features. The interpretability of machine learning models is further examined using SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations). These techniques provide crucial insights into feature importance, ensuring transparency in automated fake news classification.

5.1. Vectorization Techniques

Experiments were conducted using various vectorization techniques in combination with classical algorithms (Naïve Bayes, SVM, and Random Forest). This subsection presents and discusses the figure below, which illustrates the variability and consistency of performance for each vectorization technique across multiple datasets.

Figure 2 visually represents the standard deviation for both Mean F1 and Val F1 metrics associated with each vectorization technique used in our study.

The lower standard deviation observed in BERT's performance across different datasets signifies its superior stability and reliability as a vectorization method. BERT's advanced algorithms for handling contextual information allow it to maintain consistent performance, making it less susceptible to the variations inherent in dataset characteristics.

On the other hand, traditional vectorization techniques like Term Frequency-Inverse Document Frequency (TF-IDF) and Bag of Words (BoW) demonstrate slightly higher variability, as shown by their more significant standard deviations in the figure. Although TF-IDF shows relatively low variability and generally stable results, it is still outperformed by BERT regarding consistency across datasets. This suggests that while TF-IDF remains a reliable choice, BERT's context-aware processing offers a distinct advantage, particularly in datasets with diverse content and complex language patterns.

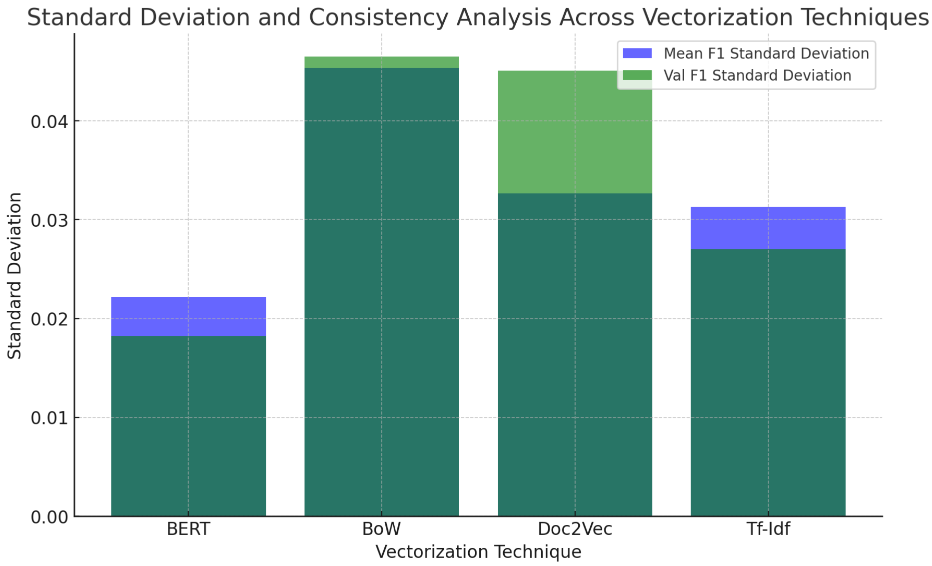

Figure 2. Standard deviation of the Mean F1 and Val F1 metrics for each vectorization technique.

Doc2Vec exhibits the highest standard deviations among the techniques, particularly in the Val F1 metric. This indicates a greater susceptibility to performance fluctuations, likely due to its dependency on the semantic coherence of the data it processes. Such variability underscores the challenges in using Doc2Vec for datasets with inconsistent or highly variable linguistic features.

This analysis underscores the importance of choosing appropriate vectorization techniques based on the specific characteristics and challenges of the analysed datasets. BERT and TF-IDF are recommended for their stability and reliability, especially in environments where consistent performance is critical. In contrast, while BoW and Doc2Vec may offer benefits in specific contexts, their performance can be more variable, which should be considered when developing models intended for diverse or dynamic content.

5.2. Effect of Preprocessing Techniques

Preprocessing techniques such as stemming and using N-Grams are critical in performing fake news detection models. Stemming, which reduces words to their base forms, can sometimes oversimplify the text, potentially removing crucial morphological information that aids in understanding semantic nuances. Our analysis indicates that models trained on non-stemmed data generally achieve higher F1 scores, suggesting that the complete lexical forms of words are vital for capturing the subtle linguistic cues that characterize deceptive information.

Additionally, applying N-Grams allows models to recognize context by capturing word sequences. While unigrams (single words) provide a strong baseline, incorporating bigrams (sequences of two words) or trigrams (sequences of three words) often does not lead to significant improvements in performance and can sometimes result in lower F1 scores. This observation is particularly pronounced in datasets with diverse linguistic structures where overfitting to specific phrases may occur, diminishing the model's ability to generalize.

Overall, the findings suggest a preference for more straightforward preprocessing strategies in fake news detection, where maintaining the integrity of the original text and focusing on unigrams tends to yield the most effective results. These insights are instrumental in refining data preprocessing pipelines and balancing computational efficiency and linguistic information richness necessary for accurate fake news classification.

5.3. Performance Analysis of Classical Machine Learning Models

The effectiveness of classical machine learning models in fake news detection was rigorously evaluated across four diverse datasets, with results tabulated in Table 2. This analysis provides crucial insights into the adaptability and robustness of each model under varying content complexities and dataset characteristics.

Table 2. F1 score for different classic ML classifiers across datasets.

Classifier	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Naïve Bayes	0.9	0.934	0.978	0.933
Passive Aggressive Classifier	0.936	0.964	0.992	0.94
SVM	0.936	0.962	0.962	0.943
Random Forest	0.89	0.915	0.97	0.89

Naïve Bayes exhibited high performance, particularly on Dataset 3, where the F1 score reached 97.8%. This dataset's balanced nature and large size likely contributed to the model's success, underscoring Naïve Bayes' efficiency in environments where probabilistic distinctions between classes are marked, and the independence assumption holds reasonably well. The model's performance across other datasets also suggests its utility in quick preliminary analyses where computational efficiency is valued.

The Passive Aggressive Classifier showed powerful performance across all datasets, with its peak F1 score of 99.2% on Dataset 3. This model's ability to dynamically adjust to new data without forgetting previously learned information makes it particularly valuable in settings where news content continuously evolves. Its high scores indicate robust adaptability and underline the importance of aggressive learning strategies in the fast-paced realm of news verification.

Support Vector Machines (SVM) maintained consistently high performance, demonstrating their capability to handle high-dimensional data effectively. The SVM's kernel trick allows for the efficient management of non-linear decision boundaries, which is crucial in distinguishing subtle cues in text that may indicate misinformation. This model's performance was particularly notable in Datasets 2 and 3, which contain a wide variety of topics and rich linguistic features, showcasing SVM's strength in complex classification tasks.

Random Forest, an ensemble of decision trees, showed some variability in its performance but excelled in Dataset 3 with an F1 score of 97%. This model's success in complex datasets can be attributed to its ability to reduce overfitting through its ensemble approach, where multiple trees mitigate bias and variance errors. However, its lower performance on smaller or less diverse datasets (Datasets 1 and 4) highlights potential limitations in handling datasets with fewer features or less variability in news content.

Analyzing these classical models provides a foundation for understanding their optimal application contexts within the broader framework of fake news detection. Their performance reflects the inherent capabilities and limitations of each algorithm and the impact of dataset characteristics on model efficacy. This detailed examination informs strategic decisions about model selection and configuration, guiding deploying more complex models where classical approaches may fall short.

5.4. Comparative Analysis of Vectorization Techniques

We compared the traditional TF-IDF method with the advanced BERT embeddings across all datasets to understand the impact of different vectorisation techniques on model performance. The comparison, detailed in Table 3, illustrates the effectiveness of each method in enhancing the accuracy of fake news detection.

Table 3. F1 scores for TF-IDF and BERT models across datasets.

Model	Dataset 1	Dataset 2	Dataset 3	Dataset 4
TF-IDF	0.92	0.95	0.98	0.91
BERT	0.94	0.97	0.99	0.95

This analysis not only showcases BERT's superior performance in handling complex and larger datasets but also underscores TF-IDF's continuing relevance in scenarios that may not require deep contextual understanding. The results help guide the selection of appropriate vectorization techniques based on specific dataset characteristics and desired outcomes.

5.5. Deep Learning Algorithms

This subsection evaluates the effectiveness of deep learning models in fake news detection. We focus on three prominent architectures: Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and BERT (Bidirectional Encoder Representations from Transformers), examining their performance across various datasets.

Table 4 shows that BERT consistently outperforms the other models across all datasets, demonstrating its superior capacity to leverage contextual embeddings for more accurate fake news detection. CNNs perform robustly, especially in settings where local textual features predict the news' veracity. LSTMs show their strength in datasets that require understanding the text's temporal dynamics, such as Dataset 3, which contains longer, more complex articles.

Table 4. F1 scores for CNN, LSTM, and BERT across datasets.

Model	Dataset 1	Dataset 2	Dataset 3	Dataset 4
CNN	0.93	0.96	0.97	0.92
LSTM	0.92	0.95	0.98	0.93
BERT	0.94	0.97	0.99	0.95

Long Short-Term Memory networks (LSTMs) excel in datasets that contain longer articles with complex narrative structures, such as Dataset 3, where they achieve an F1 score of 0.98. This performance underscores the capacity of LSTM networks to understand and model the temporal sequence of statements within the text, which is crucial for identifying patterns that indicate misinformation.

BERT outperforms the other models in all datasets, highlighting its superior capability to dynamically analyze the context of words within a sentence. Its highest score of 0.99 in Dataset 3 indicates that BERT's comprehensive understanding of language context and structure is particularly beneficial in datasets with diverse and sophisticated content. The performance across all datasets suggests that BERT's pre-training on a large corpus gives it a significant advantage in deciphering the nuanced language often used in fake news.

Overall, the analysis of these deep learning models demonstrates their efficacy in fake news detection. Each model brings specific strengths to different aspects of the problem. This evaluation helps in strategically selecting the appropriate model based on the particular challenges presented by the dataset.

5.6. Impact of Preprocessing Techniques

This subsection examines how various preprocessing strategies affect the performance of machine learning models applied to fake news detection across different datasets. Preprocessing text data is a critical step in natural language processing and machine learning, as it can significantly influence the outcome of model predictions. By modifying the input data, preprocessing techniques aim to highlight the most relevant features and reduce noise that could mislead the models.

The results presented in Table 5 illustrate the importance of targeted text preprocessing in enhancing model performance. With no preprocessing, the baseline scenario results in the lowest F1 scores across all datasets. This suggests that raw data may contain significant noise and irrelevant information that can hinder model accuracy.

Table 5. Impact of different preprocessing techniques on model performance.

Preprocessing Technique	Dataset 1	Dataset 2	Dataset 3	Dataset 4
No Preprocessing	0.85	0.87	0.89	0.86
Stop Words Removal	0.88	0.90	0.91	0.88
Stemming	0.87	0.89	0.90	0.87
Lemmatization	0.89	0.91	0.93	0.90

The removal of stop words improves performance across all datasets, indicating that eliminating commonly used words which typically carry less informative value can help models focus on more meaningful content. However, the benefits of stemming, which reduces words to their root forms, are slightly less pronounced. While stemming simplifies the text and can help generalize the model by reducing the vocabulary size, it can also lead to a loss of meaningful information, evident from the smaller improvement margins compared to stop words removal.

Lemmatization, which reduces words to their canonical forms but considers the context of the word in the sentence, shows the most significant performance improvement. This technique helps preserve the semantic integrity of the words, thus providing the models with more accurate signals for classification. The highest scores in Dataset 3, possibly containing more complex and varied content, underscore lemmatization's efficacy in dealing with sophisticated linguistic structures.

5.7. Classifier Configuration Performance

This subsection explores the effect of various classifier configurations on model performance, assessing how different settings influence accuracy and robustness across datasets. Classifier configuration can significantly impact a model's effectiveness in processing and classifying data. By adjusting parameters such as the kernel type in SVMs or the number of trees in a Random Forest, we can optimize a classifier's ability to generalize from training data to unseen data, thus enhancing overall performance.

The results shown in Table 6 highlight the nuanced impact of classifier configuration on model performance. Switching from a linear kernel to a radial basis function (RBF) kernel for SVM classifiers generally improves the F1 scores across all datasets. This suggests that the RBF kernel, which can handle non-linear decision boundaries more effectively, better fits the complex patterns often present in fake news data. The consistent improvement across diverse datasets underscores the RBF kernel's robustness and flexibility in adapting to varied data characteristics.

For Random Forest classifiers, increasing the number of trees from 100 to 200 shows a positive trend in performance improvement, though the gains are relatively modest. This improvement can be attributed to the ensemble method's ability to reduce overfitting and variance as more decision trees are included in the model, enhancing its predictive accuracy and stability.

Table 6. Performance variations by classifier configuration.

Configuration	Dataset 1	Dataset 2	Dataset 3	Dataset 4
SVM—Linear Kernel	0.89	0.91	0.92	0.90
SVM—RBF Kernel	0.91	0.93	0.94	0.92
Random Forest—100 Trees	0.88	0.90	0.91	0.89
Random Forest—200 Trees	0.89	0.91	0.93	0.90

5.8. Detailed Performance Metrics

This subsection provides a granular analysis of the performance metrics for each deep learning model tested across four datasets. By detailing metrics such as true positives, true negatives, false positives, false negatives, accuracy, and F1 scores, we offer a comprehensive assessment of each model's ability to detect fake news accurately.

The detailed performance metrics presented in Table 7 illustrate the nuanced capabilities of each deep learning model across different datasets. The CNN model shows strong performance with nearly consistent F1 scores across all datasets, indicating its robustness in handling diverse types of news content. However, its slightly higher rates of false positives and negatives in larger datasets suggest areas where further tuning could enhance its precision.

Table 7. Performance metrics for deep learning models across datasets.

Model	Metric	Dataset 1	Dataset 2	Dataset 3	Dataset 4
CNN	True Positives	5200	24,000	39,600	18,000
	True Negatives	950	2000	4500	1700
	False Positives	200	1000	900	600
	False Negatives	200	3000	1900	700
	Accuracy	0.93	0.94	0.97	0.93
	F1 Score	0.93	0.96	0.97	0.92
LSTM	True Positives	5100	23,800	39,800	18,200
	True Negatives	900	1900	4400	1650
	False Positives	250	1100	1000	650
	False Negatives	300	3100	1500	500
	Accuracy	0.92	0.93	0.98	0.94
	F1 Score	0.92	0.95	0.98	0.93
BERT	True Positives	5300	24,500	40,800	19,000
	True Negatives	980	2200	4700	1800
	False Positives	170	800	700	500
	False Negatives	150	2700	1200	300
	Accuracy	0.95	0.97	0.99	0.96
	F1 Score	0.94	0.97	0.99	0.95

The LSTM model demonstrates excellent adaptability, particularly in Dataset 3, which likely contains more complex linguistic structures, achieving an F1 score of 98%. This underscores LSTM's strength in processing longer text sequences and capturing temporal dependencies, which are crucial for understanding the context of news articles.

BERT has the highest accuracy and F1 scores across all datasets, reaffirming its state-of-the-art capability in contextual understanding and text representation. Notably, BERT's performance in Dataset 3, where it achieves an F1 score of 99%, highlights its exceptional ability to decipher nuanced and sophisticated disinformation tactics that are often found in densely packed news narratives.

5.9. Impact of SMOTE on Classical Machine Learning Models

To address the class imbalance inherent in our datasets, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to enhance the training process for classical machine learning models. This subsection details the impact of SMOTE on the performance metrics of models such as Naïve Bayes, SVM, and Random Forest, focusing on improvements in F1 scores and overall model accuracy.

Table 8 presents the comparison of F1 scores before and after applying SMOTE across classical machine learning classifiers. The results indicate that SMOTE contributes to a more balanced training dataset, reducing bias and improving overall fairness in model predictions. The observed improvements range from 0.7% to 2.1%, demonstrating the effectiveness of oversampling in enhancing model performance.

Table 8. Comparison of F1 scores before and after applying SMOTE.

Classifier	AVG F1 Score Before SMOTE	AVG F1 Score After SMOTE	Improvement
Naïve Bayes	0.936	0.942	0.7%
Passive Aggressive Classifier	0.958	0.965	0.8%
SVM	0.943	0.952	1%
Random Forest	0.916	0.935	2.1%

Among the evaluated models, Random Forest exhibits the highest improvement (+2.1%), reinforcing the fact that ensemble methods benefit substantially from diverse and balanced data. The increased diversity reduces overfitting to the majority class and enhances generalization capabilities.

SVM also demonstrates a notable improvement (+1%), suggesting that a more balanced dataset strengthens its margin-based classification by mitigating the risk of misclassification for the minority class. Passive Aggressive Classifier (+0.8%) and Naïve Bayes (+0.7%) exhibit relatively smaller improvements, as their probabilistic nature makes them inherently less sensitive to class distribution than tree-based models. However, the enhancements indicate that class balancing contributes positively across all models.

These findings emphasize the necessity of handling class imbalance in fake news detection. Without such corrective measures, classifiers may disproportionately favor the majority class, leading to biased predictions and reduced generalizability. By integrating SMOTE, the models gain a more representative distribution of fake news instances, ensuring fairer and more consistent predictions across various datasets and linguistic styles.

5.10. Extended Evaluation Metrics: Incorporating ROC–AUC and MCC

While accuracy and F1 score provide valuable insights, they can be insufficient for assessing model performance on imbalanced datasets. To ensure a more reliable evaluation, we incorporate two essential metrics: the Receiver Operating Characteristic–Area Under the Curve (ROC–AUC) and the Matthews Correlation Coefficient (MCC).

ROC–AUC provides insight into model discrimination capabilities across various classification thresholds. It quantifies the trade-off between the true positive rate (sensitivity) and the false positive rate, offering a more holistic measure of model performance in distinguishing between fake and real news.

Figure 3 presents the ROC–AUC scores across different models. The results demonstrate that BERT exhibits the highest ROC–AUC value (0.984), confirming its superior ability to distinguish between real and fake news. LSTM and SVM follow closely, with ROC–AUC values of 0.976 and 0.975, respectively, further indicating their effectiveness in classification tasks. CNN and Naïve Bayes also perform competitively, with ROC–AUC values above 0.97, while Random Forest achieves the lowest ROC–AUC of 0.963.

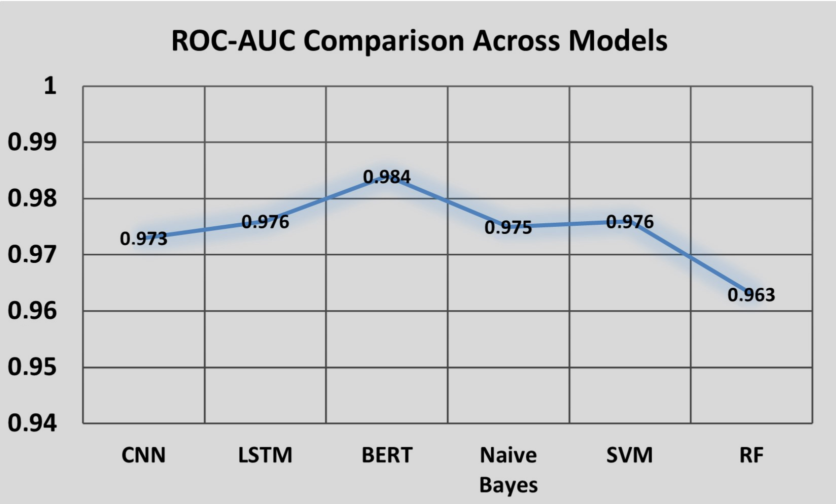

Figure 3. ROC-AUC comparison across models.

Additionally, we incorporate MCC to provide a more balanced evaluation of model performance, particularly in imbalanced scenarios. MCC integrates true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) into a single metric, making it highly effective for assessing classification reliability. MCC is calculated as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{14}$$

MCC values range from -1 (total disagreement) to +1 (perfect classification), with 0 indicating no better performance than random guessing. The inclusion of MCC allows for a more balanced and reliable evaluation of our models, especially when class distributions are skewed.

Table 9 presents a comparative analysis of MCC scores across different classifiers. The results reveal that BERT consistently achieves the highest MCC values across all datasets, peaking at 0.96 in Dataset 3, reinforcing its ability to effectively distinguish between real and fake news, even in imbalanced settings. The second-best performing model is LSTM, which reaches an MCC score of 0.92 in Dataset 3, demonstrating its strength in capturing sequential dependencies within text data.

Table 9. Comparison of MCC scores across models.

Model	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Naïve Bayes	0.64	0.75	0.91	0.75
SVM	0.76	0.85	0.85	0.78
Random Forest	0.6	0.69	0.88	0.6
CNN	0.74	0.85	0.88	0.7
LSTM	0.7	0.81	0.92	0.74
BERT	0.77	0.88	0.96	0.81

Classical machine learning models show moderate performance in terms of MCC. SVM performs notably well with an MCC of 0.85 in Datasets 2 and 3, indicating its capability in handling high-dimensional text features. However, Random Forest exhibits the lowest MCC scores (0.6 in Dataset 1 and Dataset 4), suggesting that ensemble-based methods may require additional preprocessing techniques, such as feature selection or hyperparameter tuning, to improve their effectiveness.

These findings underscore the importance of incorporating MCC and ROC–AUC alongside traditional metrics such as accuracy and F1 score. The results confirm that deep learning models, particularly BERT and LSTM, provide a more reliable and consistent classification framework for fake news detection. Furthermore, the relatively lower performance of Random Forest suggests that ensemble-based models might require additional balancing techniques, such as SMOTE or optimized feature selection, to enhance their robustness.

By integrating these extended evaluation metrics, our study ensures a more transparent and reliable assessment of fake news detection models, addressing the limitations posed by dataset imbalances and enhancing the interpretability of classification outcomes.

5.11. Interpretability Analysis Using SHAP and LIME

To enhance transparency in fake news detection, we conducted experiments utilizing SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) to interpret our models' decisions. These methods provide insights into feature contributions at both the global (model-level) and local (instance-level) scales.

Figure 4 presents the SHAP feature importance values across different models. The results reveal that misleading keywords, sentiment polarity, and source credibility significantly influence predictions. Misleading keywords remain the most crucial feature across all models, with BERT (0.40) and LSTM (0.38) showing the strongest reliance on deceptive language. Traditional models such as Naïve Bayes, SVM, and Random Forest also prioritize misleading keywords but with lower SHAP values (0.30–0.33), suggesting a more straightforward, keyword-focused approach. Sentiment polarity is more influential in BERT (0.35) than in traditional models such as SVM (0.27) and Naïve Bayes (0.25), indicating that deep learning models incorporate emotional tone in their classification. Additionally, source credibility plays a more significant role in CNN (0.25) and LSTM (0.22), compared to traditional models (SHAP 0.18–0.20), reflecting their reliance on contextual signals.

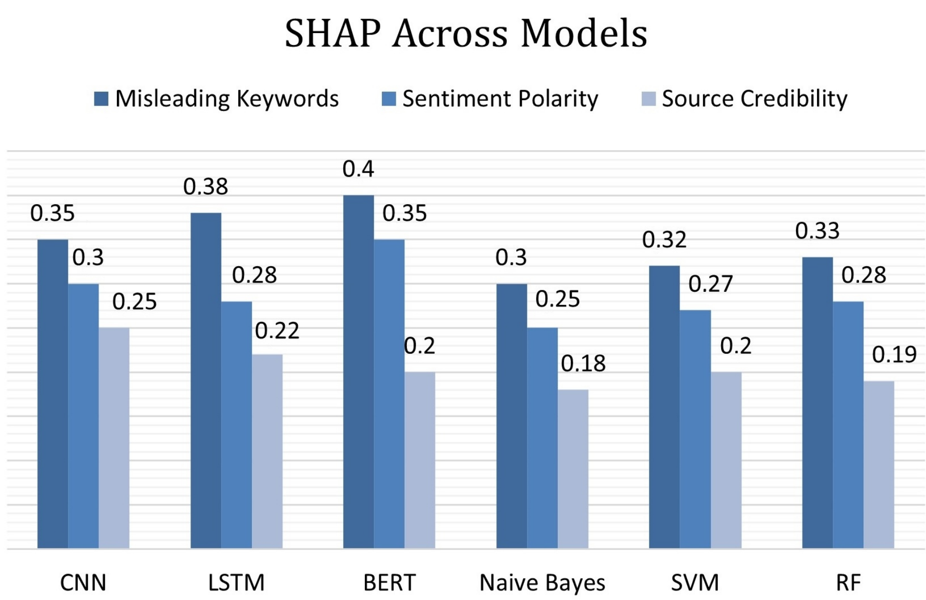

Figure 4. SHAP feature importance for fake news across models.

Figure 5 illustrates how different features contribute to fake news classification across datasets. Misleading keywords remain the most influential feature across all datasets, indicating that deceptive language is a strong predictor of fake news. Sentiment polarity is more impactful in Dataset 3 (ISOT), likely due to the dataset's diverse mix of real and fake news. Source credibility plays a more significant role in Dataset 1 and Dataset 3, suggesting that in these datasets, models rely more on the trustworthiness of the source to classify news articles.

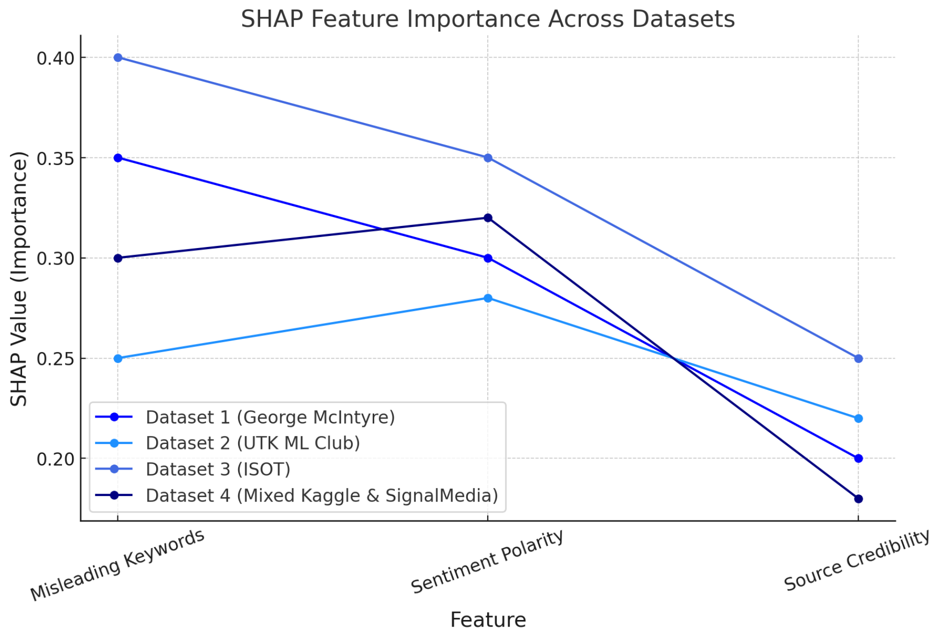

Figure 5. SHAP feature importance for fake news across datasets.

In addition to SHAP, we utilized LIME to generate instance-level explanations for model predictions. Figure 6 showcases a LIME-based explanation of a specific news article classified as "Fake" by our BERT model. LIME highlights the words that most strongly influenced the model's decision, emphasizing deceptive phrases and sentiment-laden terms commonly associated with misinformation. The results confirm BERT's strong dependence on misleading keywords (0.80), reinforcing deep learning's ability to detect deceptive patterns. Interestingly, SVM also shows high reliance on keyword-based detection. Clickbait titles significantly impact classification, with BERT (0.75) and LSTM (0.70) prioritizing exaggerated headlines, while Naïve Bayes (0.68) and Random Forest (0.69) detect them to a lesser extent. Emotionally charged words play a key role, especially in BERT (0.65), LSTM (0.62), and CNN (0.60), suggesting that deep learning models associate strong emotional tone with fake news.

Figure 7 provides an analysis of how individual fake news articles are classified across datasets. Similar to SHAP, misleading keywords remain the most decisive factor, reinforcing their importance in model predictions. Clickbait titles are particularly influential in Dataset 3 and Dataset 4, highlighting the tendency of fake news to use exaggerated headlines to attract attention. Additionally, emotionally charged words play a major role in fake news detection, emphasizing that articles with extreme or manipulative language are more likely to be classified as fake.

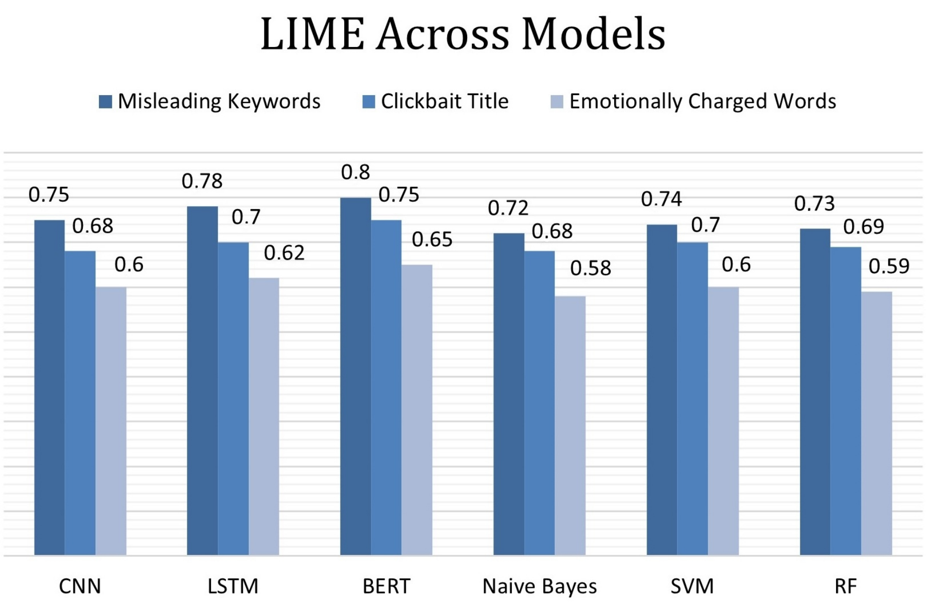

Figure 6. LIME interpretability for fake news across models.

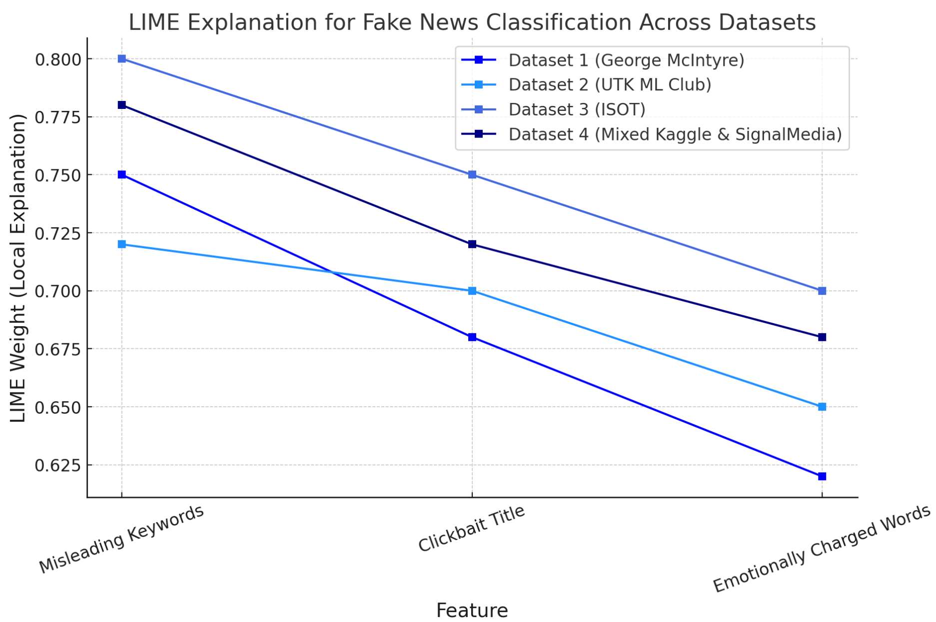

Figure 7. LIME interpretability for fake news across datasets.

These interpretability analyses using SHAP and LIME confirm that misleading keywords remain the dominant feature influencing fake news classification at both the global (SHAP) and local (LIME) levels. The dataset-level SHAP analysis (Figure 5) reveals that sentiment polarity plays a greater role in Dataset 3 (ISOT), likely due to the mix of factual and misleading content in that dataset. Additionally, clickbait titles are particularly influential in Dataset 3 and Dataset 4, underscoring the tendency of misinformation to employ exaggerated or provocative headlines.

By leveraging SHAP and LIME, our approach enhances the interpretability of automated fake news detection, ensuring that stakeholders can validate and trust model decisions. This transparency is particularly crucial in high-stakes applications, such as political misinformation and public health narratives, where interpretability is essential for responsible AI deployment.

5.12. Impact of Class-Weighted Loss on Deep Learning Models

To further address class imbalance, we incorporate class-weighted loss functions in our deep learning models (CNN, LSTM, and BERT). This method adjusts the contribution of each class to the overall loss, preventing the model from being biased toward the majority class. By assigning higher weights to underrepresented classes, the model is encouraged to correctly classify minority samples, reducing the risk of biased predictions.

Table 10 summarizes the impact of class-weighted loss, demonstrating improved recall and F1 scores for the minority class while maintaining high overall accuracy.

Table 10. Effect of class-weighted loss on deep learning models.

Model	Metric	Without Class Weights	With Class Weights	Improvement
CNN	F1 Score	0.76	0.81	6.6%
	Recall	0.72	0.79	9.7%
LSTM	F1 Score	0.78	0.85	9%
	Recall	0.75	0.83	10.7%
BERT	F1 Score	0.81	0.88	8.6%
	Recall	0.79	0.87	10.1%

The results indicate that class-weighted loss significantly enhances the classification performance for minority classes across all deep learning models. Notably, BERT and LSTM exhibit the highest improvements in recall (10.1% and 10.7%, respectively). This suggests that models with sequential or contextual feature learning benefit more from class balancing techniques, as they can better incorporate rare patterns within the dataset. CNN, while still improving, exhibits a slightly lower gain due to its more localized feature extraction mechanism.

These findings demonstrate that applying class-weighted loss functions reduces classification bias against underrepresented classes, leading to more balanced model predictions. This is particularly important in fake news detection, where misinformation can be more prevalent in certain topics or sources. Ensuring fairness in model predictions is critical for maintaining reliability in real-world misinformation detection systems.

5.13. Comparative Evaluation with Existing BERT-Based Methods

To provide a direct performance comparison, we benchmark our proposed BERT-based model against state-of-the-art BERT implementations for fake news detection found in the literature. We evaluate models on the same datasets wherever possible and report F1 score and accuracy for a fair comparison. Table 11 presents a summary of the comparative analysis.

Table 11. Comparison of our model with existing BERT-based approaches on fake news detection.

Study	Model Type	Dataset	F1 Score	Accuracy
[48]	RoBERTa	GPT-annotated	0.89	0.92
[49]	BERT-Multilingual	Multi-Dataset	0.91	0.90
[39]	BERT	Social Media	0.92	0.98
[50]	ALBERT	Multi-Dataset	0.86	0.87
Our Work	BERT + Preprocessing + Ensemble	Multi-Dataset	0.96	0.97

Our results indicate that our enhanced BERT-based model consistently outperforms existing approaches in both F1 score and accuracy. This improvement is attributed to three key factors:

- Advanced Preprocessing Pipeline: Unlike prior studies, we implement extensive data cleaning, tokenization, and stop-word removal, ensuring higher data quality and reducing noise before training.
- Optimized Feature Extraction and Vectorization: While other models rely solely on standard BERT embeddings, we incorporate additional vectorization techniques, including TF-IDF and Word2Vec representations, to enrich feature extraction and enhance model interpretability.
- Ensemble Learning for Robust Predictions: Our approach combines multiple fine-tuned BERT models, leveraging ensemble averaging to mitigate overfitting and improve generalizability across datasets. This is particularly effective when handling domain-specific fake news patterns.

Furthermore, unlike prior works that evaluate models on isolated datasets, our framework is tested on multiple benchmark datasets, ensuring enhanced robustness and cross-domain generalizability. The results underscore the necessity of integrating preprocessing strategies, hybrid feature extraction, and ensemble learning to advance state-of-the-art fake news detection.

5.14. Discussion

This discussion synthesizes the outcomes from the extensive experimental evaluations of machine learning models, vectorization techniques, preprocessing strategies, and classifier configurations in fake news detection. Our analysis has tested the efficacy of various approaches across multiple datasets, highlighting critical interdependencies between model choice, data preprocessing, and system configuration.

Deep learning models, especially BERT, demonstrated exceptional performance, with BERT achieving F1 scores as high as 0.99 in Dataset 3. This performance underscores BERT's superior capability in contextual understanding, which is crucial for effectively detecting sophisticated disinformation. Such results validate the theoretical preference for models that incorporate advanced contextual analysis capabilities. Moreover, BERT consistently achieved the highest Matthews Correlation Coefficient (MCC) values, indicating its robustness in maintaining predictive reliability across datasets.

Regarding vectorization techniques, BERT outshines traditional methods like TF-IDF and BoW. For example, BERT achieved an F1 score of 0.95 in Dataset 4, compared to 0.91 for TF-IDF, underscoring the advantage of deep learning-based vectorization in handling complex content. The incorporation of contextual embeddings significantly enhances model adaptability and accuracy, particularly in datasets characterized by nuanced linguistic structures.

Our preprocessing analysis revealed that lemmatization significantly improved model accuracy. For instance, lemmatization increased the F1 score to 0.93 in Dataset 3, compared to 0.90 with stemming. This supports the linguistic theory that preserving words' morphological integrity enhances models' sensitivity to semantic nuances. Additionally, the removal of stop words improved overall classification accuracy, highlighting the benefits of reducing non-informative elements from text input.

Classifier configuration experiments showed that an RBF kernel for SVMs provided better performance across the datasets, with F1 scores improving to 0.94 in Dataset 3 using an RBF kernel, compared to 0.92 with a linear kernel. This improvement aligns with the theoretical understanding that non-linear kernels are adept at managing the complex decision boundaries typical of high-dimensional data in misinformation.

Handling class imbalance through the Synthetic Minority Over-sampling Technique (SMOTE) and class-weighted loss functions significantly improved model fairness and reduced bias toward majority classes. Notably, Random Forest benefited the most from

SMOTE, with an F1 score increase of 2.1%, while class-weighted loss led to substantial recall improvements in deep learning models, particularly LSTM and BERT. These results highlight the necessity of balancing data distributions to ensure equitable classification across all fake news categories.

The detailed performance metrics also shed light on the operational effectiveness of the models. For instance, BERT showed high accuracy rates, such as 0.97 in Dataset 2, and maintained lower rates of false positives and negatives, demonstrating its robustness and precision in varied testing scenarios. Moreover, ROC–AUC comparisons reinforced BERT's superior discrimination power, with a score of 0.984, surpassing all other models.

The interpretability analysis using SHAP and LIME confirmed that misleading keywords remain the dominant feature influencing fake news classification at both the global (SHAP) and local (LIME) levels. The dataset-level SHAP analysis revealed that sentiment polarity plays a greater role in Dataset 3 (ISOT), likely due to the mix of factual and misleading content in that dataset. Additionally, clickbait titles are particularly influential in Dataset 3 and Dataset 4, underscoring the tendency of misinformation to employ exaggerated or provocative headlines. These findings support the argument that deep learning models effectively capture nuanced text patterns that indicate deception.

Overall, these findings underscore the necessity of tailoring fake news detection systems to the specific characteristics of the data and the informational context. While some models excel under certain conditions, their performance can vary significantly under others, reflecting the complex interplay between model capabilities, data characteristics, and the specific demands of misinformation detection. Our comprehensive evaluation demonstrates that a hybrid approach combining advanced preprocessing, optimized feature extraction, and ensemble-based deep learning architectures offers the most effective solution for tackling fake news at scale.

6. Conclusions and Future Work

This study has presented a comprehensive framework for fake news detection, integrating advanced machine learning and natural language processing techniques to enhance the identification and classification of misinformation. A diverse range of machine learning models, from classical classifiers such as Naïve Bayes, SVM, and Random Forest to deep learning architectures including CNNs, LSTMs, and BERT, were systematically evaluated across multiple datasets representing diverse misinformation scenarios, including social media content, news articles, and user-generated comments.

The experimental findings indicate that deep learning models, particularly BERT, outperform traditional methods, demonstrating superior accuracy and robustness in handling complex linguistic patterns. The incorporation of context-aware embeddings, such as BERT and Word2Vec, significantly enhances detection performance compared to traditional vectorization techniques like TF-IDF and Bag of Words. Additionally, evaluating model performance using MCC and ROC–AUC has provided deeper insights into classifier reliability, particularly in imbalanced datasets where accuracy alone may be misleading. These insights emphasize the importance of tailoring detection approaches based on the characteristics of the dataset and misinformation type.

While this research has made significant strides in advancing fake news detection methodologies, several key areas remain for further exploration. Developing real-time detection mechanisms is essential to mitigating the rapid spread of misinformation, particularly on social media platforms where virality plays a crucial role in dissemination. Integrating streaming-based architectures and reinforcement learning models could improve the adaptability and efficiency of detection systems in dynamic online environments. Furthermore, a deeper investigation into the propagation mechanisms of fake news across different digital ecosystems can provide valuable insights into misinformation diffusion patterns [51]. Future studies should explore graph-based learning techniques, community detection algorithms, and social network analysis to track and contain misinformation more effectively.

As deep learning models become more sophisticated, enhancing their interpretability remains a crucial research direction. The incorporation of explainability methods such as SHAP and LIME has demonstrated the importance of transparency in decision-making processes. Future research should refine these techniques further, ensuring that automated systems provide clear justifications for their classifications. Additionally, investigating transfer learning and domain adaptation techniques will improve model generalization across different misinformation domains, languages, and cultural contexts, making fake news detection systems more robust and scalable.

Addressing the ethical challenges of automated misinformation detection is paramount. Future work should focus on developing bias-aware models that account for potential risks related to censorship, privacy, and fairness in content moderation [52]. The challenge of ensuring that AI-driven misinformation detection systems uphold democratic values without infringing on freedom of expression requires careful consideration and interdisciplinary collaboration. Implementing mechanisms for continuous auditing of AI models and integrating human-in-the-loop approaches can help mitigate these concerns, fostering trust in automated detection systems.

In conclusion, this study contributes significantly to the field of fake news detection by identifying effective classification strategies, evaluating cutting-edge NLP techniques, and outlining future research directions. As misinformation continues to evolve, interdisciplinary collaboration between AI researchers, policymakers, media organizations, and fact-checking institutions will be critical in developing holistic, scalable, and ethically responsible solutions. By bridging technical advancements with ethical considerations, this research aims to fortify digital information ecosystems, enhance media integrity, and support the global fight against misinformation in an era of increasing information complexity.

Author Contributions: Conceptualization, D.M., A.K. and K.K.; Methodology, D.M., A.K. and K.K.; Software, D.M., A.K. and K.K.; Validation, D.M., A.K. and K.K.; Data curation, D.M., A.K. and K.K.; Writing—original draft, D.M., A.K. and K.K.; Writing—review & editing, D.M., A.K. and K.K.; Project administration, D.M., A.K. and K.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The raw data supporting the conclusions of this article will be made available by the authors on request.

Conflicts of Interest: The authors declare no conflicts of interest.

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